Geometric transformations

Geometric transformations can be defined as mappings of points from one space to another, with emphasis on **linear transformations** that preserve co-linearity, ratios of distances, and parallelism, and that map the origin to itself.

- The five fundamental transformations are: **translation, scaling, rotation, reflection, and shearing**.

Transformation	2D Matrix	Inverse 2D Matrix
Translation	$\begin{bmatrix}1 & 0 & t_x\\0 & 1 & t_y\\0 & 0 & 1\end{bmatrix}$	$\begin{bmatrix}1 & 0 & -t_x\\0 & 1 & -t_y\\0 & 0 & 1\end{bmatrix}$
Scaling	$\begin{bmatrix}s_x & 0 & 0\\0 & s_y & 0\\0 & 0 & 1\end{bmatrix}$	$\begin{bmatrix}1/s_x & 0 & 0\\0 & 1/s_y & 0\\0 & 0 & 1\end{bmatrix}$
Reflection	$horizontal\ reflection: \begin{bmatrix}1 & 0 & 0\\0 & -1 & 0\\0 & 0 & 1\end{bmatrix}$ $vertical\ reflection: \begin{bmatrix}-1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$	$horizontal\ reflection: \begin{bmatrix}1 & 0 & 0\\0 & -1 & 0\\0 & 0 & 1\end{bmatrix}$ $vertical\ reflection: \begin{bmatrix}-1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$
Rotation	$R(\theta) = \begin{bmatrix}cos\,\theta & -sin\,\theta & 0\\sin\,\theta & cos\,\theta & 0\\0 & 0 & 1\end{bmatrix}$	$R^T(\theta) = \begin{bmatrix}cos\,\theta & sin\,\theta & 0\\-sin\,\theta & cos\,\theta & 0\\0 & 0 & 1\end{bmatrix}$
Shearing	$shear\ in\ y: \begin{bmatrix}1 & 0 & 0\\sh_y & 1 & 0\\0 & 0 & 1\end{bmatrix}$ $shear\ in\ x: \begin{bmatrix}1 & sh_x & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$	$shear\ in\ y: \begin{bmatrix}1 & 0 & 0\\1/sh_y & 1 & 0\\0 & 0 & 1\end{bmatrix}$ $shear\ in\ x: \begin{bmatrix}1 & 1/sh_x & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$

Transformation	3D Matrix	Inverse 3D Matrix
Translation	$\begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & -t_x \\ 0 & 1 & 0 & -t_y \\ 0 & 0 & 1 & -t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Scaling	$\begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1/s_x & 0 & 0 & 0 \\ 0 & 1/s_y & 0 & 0 \\ 0 & 0 & 1/s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Reflection	$yz\ plan\ reflection:\ \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $xz\ plan\ reflection:\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $xy\ plan\ \ reflection:\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$yz\ plan\ reflection:\ \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $xz\ plan\ reflection:\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $xy\ plan\ \ reflection:\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Rotation Need to specify which axis the rotation is about**.**	$Rx(\theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & cos\,\theta & -sin\,\theta & 0 \\ 0 & sin\,\theta & cos\,\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $Ry(\theta) = \begin{bmatrix} cos\,\theta & 0 & sin\,\theta & 0 \\ 0 & 1 & 0 & 0 \\ -sin\,\theta & 0 & cos\,\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $Rz(\theta) = \begin{bmatrix} cos\,\theta & -sin\,\theta & 0 & 0 \\ sin\,\theta & cos\,\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$Rx^T(\theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & cos\,\theta & sin\,\theta & 0 \\ 0 & -sin\,\theta & cos\,\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $Ry^T(\theta) = \begin{bmatrix} cos\,\theta & 0 & -sin\,\theta & 0 \\ 0 & 1 & 0 & 0 \\ sin\,\theta & 0 & cos\,\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $Rz^T(\theta) = \begin{bmatrix} cos\,\theta & sin\,\theta & 0 & 0 \\ -sin\,\theta & cos\,\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Solve

- A triangle is defined by three points A(2,2), B(3,5), c(4,4) what's the transformation coordinate after horizontal reflection.

  Answer: A(2,-2), B(3,-5), c(4,-4)

- Rotate Point p by angle 50 degree, the rotate it by angle -60 then scale it to half vertically

  Answer

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} \cos -10 & -\sin -10 & 0 \\ \sin -10 & \cos -10 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Rotate point p by angle 20 around point (5,6) then rotate angle by -50 around point (3,7)

  Answer

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} \cos 40 & -\sin 40 & 0 \\ \sin 40 & \cos 40 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} \cos 20 & -\sin 20 & 0 \\ \sin 20 & \cos 20 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 0 & 0 & -5 \\ 0 & 0 & -6 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- Magnify a triangle with vertices A(1,1), B(2, 3) and c(7, 3) to twice its size while keeping c(7,3) fixed

  Answer

$$\mathbf{P_A'} = \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 & 0 & -7 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\mathbf{P_B'} = \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 & 0 & -7 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

$$\mathbf{P_C'} = \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 & 0 & -7 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 7 \\ 3 \\ 1 \end{bmatrix}$$

- For A pyramid centred at (5,2,1) do
  - Scale in object space by 2 in the x , y, and 3 in the z
  - Rotate clockwise in object's space by 10º around x axis, 20º around y, and 70º around z
  - Translate by (1,7,3) in world space

  Answer:

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 0 & 0 & 6 \\ 0 & 1 & 0 & 9 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} cos\,70 & -sin\,70 & 0 & 0 \\ sin\,70 & cos\,70 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} cos\,20 & 0 & sin\,20 & 0 \\ 0 & 1 & 0 & 0 \\ -sin\,20 & 0 & cos\,20 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & cos\,10 & -sin\,10 & 0 \\ 0 & sin\,10 & cos\,10 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$